# H4 environmental-variable selection

Step 4: outcome-blind PC1 representative per category (max |PC1 loading|; Jolliffe 1972). Heat terms kept for the four equations (MRT dropped).

| | UCM | | | Atmo raw | Atmo average | LYS1 raw | LYS2 raw | LYS average | GVI imagery | SVF imagery | Sound | Solar | Empatica | Eye-tracker | Questionnaire | Metadata / QC | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **STEP 1**<br>**Inventory (208)** | AH<br>AIR RH<br>AIR T bot<br>AIR T mid<br>AIR T top<br>AIR temp<br>fp<br>GPS speed kmh<br>HUMIDEX<br>HUMIDEX SR<br>IO flag<br>IR back<br>IR down<br>IR front | IR left<br>IR right<br>IR spot gnd<br>IR spot left<br>IR spot right<br>IR spot sky<br>IR up<br>MRT<br>MRT S<br>OPT<br>PET<br>PET SR<br>SND dBA<br>SUN alt | SUN az<br>SUN Gh<br>UTCI<br>UTCI SR<br>WIND AWA<br>WIND AWS<br>WIND TWD<br>WIND TWS | left humidity<br>left pm1<br>left pm10<br>left pm2.5<br>left temperature<br>right humidity<br>right pm1<br>right pm10<br>right pm2.5<br>right temperature | pm1 avg<br>pm10 avg<br>pm2.5 avg | LYS1 lux cat<br>LYS1 b'<br>LYS1 g'<br>LYS1 kelvin<br>LYS1 lux<br>LYS1 medi<br>LYS1 movement<br>LYS1 r'<br>LYS1 rgbb<br>LYS1 rgbg<br>LYS1 rgbir<br>LYS1 rgbr | LYS2 lux cat<br>LYS2 b'<br>LYS2 g'<br>LYS2 kelvin<br>LYS2 lux<br>LYS2 medi<br>LYS2 movement<br>LYS2 r'<br>LYS2 rgbb<br>LYS2 rgbg<br>LYS2 rgbir<br>LYS2 rgbr | kelvin avg<br>lux avg<br>medi avg | Grass phase mean<br>GVI 50m<br>GVI grass 50m<br>GVI phase mean<br>GVI plant 50m<br>GVI tree 50m<br>Plant phase mean<br>Tree phase mean | Building %<br>Combined Vegetation %<br>Grass %<br>Plant %<br>Sky %<br>SVF<br>Tree %<br>Wall % | band proportion 1000-4000 Hz<br>band proportion 125-1000 Hz<br>band proportion 20-125 Hz<br>band proportion 4000-8000 Hz<br>band proportion 8000-20000 Hz<br>dominant frequency (Hz)<br>octave proportion 1000 Hz<br>octave proportion 125 Hz<br>octave proportion 2000 Hz<br>octave proportion 250 Hz<br>octave proportion 31.5 Hz<br>octave proportion 4000 Hz<br>octave proportion 500 Hz<br>octave proportion 63 Hz | octave proportion 8000 Hz<br>spectral centroid (Hz)<br>spectral rolloff 85% (Hz) | Clearness Index<br>Clearness Index G0<br>Clearness Index Infer | acc x<br>acc y<br>acc z<br>eda phasic<br>eda scl usiemens<br>eda tonic<br>heart rate<br>hrv fd hf<br>hrv fd hfnu<br>hrv fd lf<br>hrv fd lf hf ratio<br>hrv fd lfnu<br>hrv fd total power<br>hrv fd vlf | hrv td rmssd<br>hrv td sdnn<br>met<br>skin temperature<br>vector magnitude | blink duration s<br>distance from center<br>eyelid aperture avg<br>fixation duration s<br>fixation rate<br>gaze centrality<br>gaze dispersion<br>gaze velocity<br>pupil change rate<br>pupil diameter avg<br>saccade amplitude<br>saccade duration s<br>saccade peak velocity<br>stress composite | current comfort 1 7<br>environmental comfort mean 1 5<br>pleasantness 1 5<br>prs11 being away mean 1 5<br>prs11 coherence mean 1 5<br>prs11 fascination mean 1 5<br>prs11 scope mean 1 5<br>prs11 total mean 1 5<br>space familiarity 1 5<br>stai6 mean 1 5<br>thermal comfort 1 5 | age<br>bike frequency<br>bmi category who<br>bmi kg m2<br>clothing accessories<br>clothing footwear<br>clothing height raw<br>clothing lower body<br>clothing skin tone<br>clothing upper body<br>clothing weight raw<br>commute modes<br>daily walking<br>Date | Date (Atmo/LYS)<br>Date (eye)<br>Date (UCM)<br>Datetime<br>gender<br>GPS alt<br>GPS hdg<br>GPS HDOP<br>GPS lat<br>GPS lon<br>GPS speed valid sample count<br>GVI match dist m<br>has children<br>height m | HRV 5min available<br>HRV 5min max gap (s)<br>HRV 5min removed pct<br>HRV 5min segment end<br>HRV 5min segment start<br>HRV 5min valid rri count<br>HRV 5min valid span (s)<br>MAG hdg<br>name<br>nationality<br>own bike for experiment<br>participant id<br>ParticipantID<br>pct complete | phase start offset s<br>PhaseID<br>regular physical activity<br>residence type<br>residential area<br>response index<br>site familiarity 1 10<br>vision correction<br>weight kg<br>years in brussels |
| **STEP 2**<br>**Representatives (15)**<br>**Rules + coverage** | AIR RH<br>AIR temp<br>HUMIDEX | MRT<br>PET<br>SND dBA | SUN Gh<br>UTCI<br>WIND AWS | | pm1 avg<br>pm10 avg<br>pm2.5 avg | | | kelvin avg<br>lux avg | GVI 50m | SVF | | | | | | | | | |

| | Thermal | Index | Acoustic | Radiation | Pollution | Light | Greenery |
|---|---|---|---|---|---|---|---|
| **STEP 3**<br>**Domain Grouping (15)** | AIR temp<br>AIR RH<br>WIND AWS | HUMIDEX<br>PET<br>UTCI | SND dBA | MRT<br>SUN Gh<br>SVF | pm1 avg<br>pm2.5 avg<br>pm10 avg | lux avg<br>kelvin avg | GVI 50m |
| **STEP 4**<br>**PC1 representative per category**<br>**(+ heat terms)** | AIR temp<br>AIR RH<br>WIND AWS | HUMIDEX<br>PET<br>UTCI | SND dBA | SUN Gh<br>PC1 tie: SUN Gh vs MRT; tie-break: larger within-person SD (213 vs 3.79) | pm2.5 avg | lux avg<br>PC1 tie: lux avg vs kelvin avg; tie-break: larger within-person SD (5300 vs 601) | GVI 50m |

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| **STEP 5**<br>**Within-person term grid**<br>**(4 equations; grouped term order)** | sound + (humidity x sound) + (air temperature x sound) + (apparent wind x sound) \| global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) \| air temperature + humidity + apparent wind + Phase | max VIF = 2.21 |
| | sound + (HUMIDEX x sound) \| global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) \| HUMIDEX + Phase | max VIF = 1.16 |
| | sound + (PET x sound) \| global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) \| PET + Phase | max VIF = 2.25 |
| | sound + (UTCI x sound) \| global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) \| UTCI + Phase | max VIF = 2.45 |
| | Phase: sum-to-zero coding of 4 phases (WalkG, WalkU, BikeG, BikeU); each compared with the mean of all phases, no reference phase. | |